Electrochemical Treatment of Model Amine-Rich Wastewater

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Objective and contribution

Amine-based CO_2 capture generates amine-rich wastewater (ARWW) that requires treatment alongside solvent regeneration. The project asked which operating conditions favored removal of monoethanolamine (MEA) from model wastewater and what pathways could explain its decrease. I designed and operated a batch reactor combining electrooxidation, electrocoagulation, and granular activated carbon (GAC) adsorption, and monitored MEA removal and nitrogen species. I combined condition-specific removal trends, literature evidence, and electrode-cost considerations to recommend operating choices and examine possible removal pathways.

Experimental approach

The initial screening used 500 mL of model wastewater in a stirred batch reactor with sodium persulfate addition. The combined system was intended to couple electrochemical transformation with capture by coagulation and adsorption. I varied stainless steel (SUS), Ti, and Al electrodes, voltage, spacing, GAC dosage, influent MEA concentration, and temperature one at a time. MEA concentration was estimated by acid titration of samples taken over 120 min. The initial electrode and GAC comparisons used a pH of 10.7.

Operating factors and condition selection

The initial single-run screening findings for six factors are summarized below. Conditions were varied individually, so the retained settings were working choices within the tested ranges.

Table 1. Initial single-run screening and working-condition choices.

Factor	Initial screening finding	Working choice or implication
Electrode material	Highest observed removal with SUS; Al was close.	Al, considering performance and cost
GAC dosage	Removal increased over 10–50 g per 500 mL.	50 g within the tested range.
Initial MEA concentration	Higher percentage removal at greater dilution.	A mid-range feed for further screening.
Voltage	Highest observed removal at 30 V.	30 V.
Electrode spacing	Smaller response differences over 1–5 cm.	1 cm for further screening.
Temperature	Reported removal increased over 25–90 °C.	Priority for further evaluation.

Selected screening comparisons

Figure 1 illustrates two choices within this screening. At 120 min, reported removal was 69.6% with SUS, 60.8% with Ti, and 67.3% with Al; I retained Al after considering its lower electrode cost and literature on electrode behavior. GAC dosages of 10, 25, and 50 g per 500 mL yielded 11.1%, 36.8%, and 67.3% removal, respectively, supporting 50 g as a working dose within the tested range.

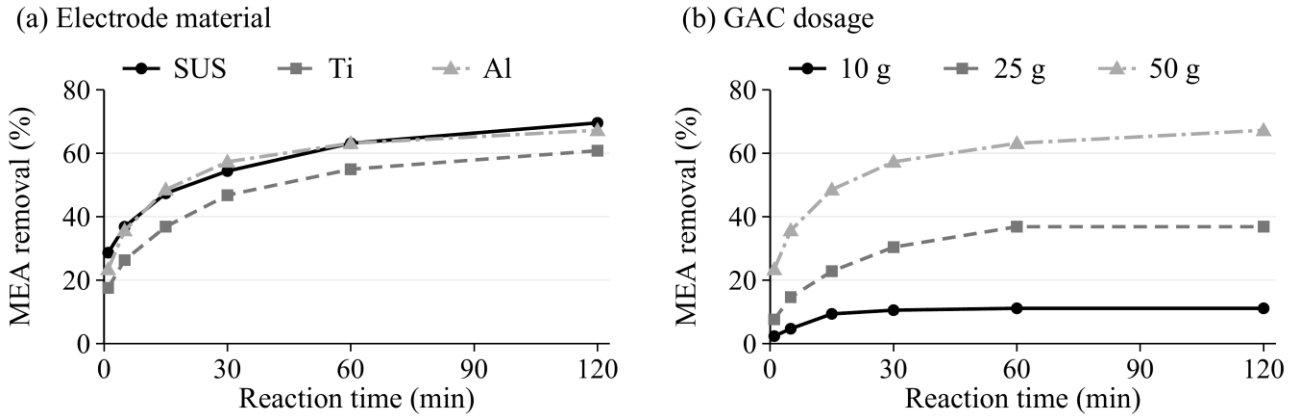


Figure 1. MEA removal during initial single-run screening. (a) Electrode material; (b) GAC dosage. Tests used 500 mL of model wastewater at initial pH 10.7. These panels precede the follow-up statistical analysis.

Follow-up experiments and operating guidelines

After the initial screening, I conducted follow-up experiments and used ANOVA with Tukey post hoc comparisons to assess operating factors. Electrode material, GAC dosage, and temperature emerged as priorities for operating guidelines within the tested system. The panels in Figure 1 document only the earlier single-run screening.

Nitrogen observations and possible removal pathways

Ammonium nitrogen was detected, while nitrate nitrogen was not detected. I interpreted these observations alongside the alkaline conditions and literature evidence to propose oxidation followed by possible ammonia release, and capture of intermediate products through coagulation and adsorption.

The study did not directly track degradation products, close a nitrogen balance, or independently quantify each pathway. Evaluation of actual capture-process wastewater remains a further step. Within the tested MEA model system, the work connected condition comparisons, nitrogen observations, and literature evidence to operating choices and testable pathway hypotheses.